

XUANFEI REN

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Education

University of Wisconsin-Madison

PhD student, advised by [Tengyang Xie](#), Department of Computer Sciences.

Madison, USA

Aug. 2024 – Present

University of Science and Technology of China

Bachelor of Science in Mathematics, specialization in Probability and Statistics.

Hefei, China

Aug. 2020 – Jun. 2024

Experience

Amazon Web Services, Annapurna Labs

Applied Scientist Intern, Neuron Science team, working with Shengzhe Xu and Shaowei Zhu.

Cupertino, USA

May 2026 – Present

- Working on **agentic kernel optimization** for AWS Trainium AI accelerators.

Research Interests

My research interests span **AI agents** and **reinforcement learning**, encompassing both theoretical foundations and experimental work.

Publication

- **When Does Trajectory-Level Supervision Permit Efficient Offline Reinforcement Learning?**
[Xuanfei Ren](#), Tengyang Xie. *Submitted to NeurIPS 2026.*
- **POLCA: Stochastic Generative Optimization with LLM**
[Xuanfei Ren](#), Allen Nie, Tengyang Xie, Ching-An Cheng. *ICML 2026.*
- **Optimal Batched Linear Bandits**
[Xuanfei Ren](#), Tianyuan Jin, Pan Xu. *ICML 2024.*

Research Projects

Outcome-Based Offline Reinforcement Learning

Work with Prof. [Tengyang Xie](#)

- Developed a statistical theory for **offline RL from trajectory-level supervision**: proposed the **OPAC** algorithm, which jointly learns a latent per-step reward model and optimizes policies via pessimistic actor-critic updates; proved a sharp guarantee of order $\tilde{O}(H^2 \sqrt{C_{\text{sa}}(\pi^*)/n})$ with a matching lower bound; extended the framework to **trajectory-level preference feedback**.
- Introduced a **generalized outcome-based RL** formulation, proving an exponential lower bound for all-success objectives and identifying tractable regimes via the structural coefficients.

Stochastic Generative Optimization with LLM

Work with Dr. [Ching-An Cheng](#) (Google Research), Dr. [Allen Nie](#) (Google DeepMind), and Prof. [Tengyang Xie](#)

- Formalized complex system tuning as a **stochastic generative optimization** problem, utilizing LLMs as optimizers to automate iterative refinement through numerical rewards and natural language feedback.
- Engineered a framework combining a **priority queue-based search strategy** to balance exploration and exploitation, an **ϵ -Net mechanism** to ensure parameter diversity, and an **LLM Summarizer** for meta-learning, achieving **provable scalability** over naive baselines under noisy evaluations.
- Demonstrated state-of-the-art performance across diverse benchmarks: **τ -bench** (agent orchestration), **VeriBench** (code translation/Lean4), and **KernelBench** (high-performance CUDA kernels).
- Contributed to [OpenTrace](#), an open-source library for end-to-end generative workflow optimization.

Optimal Batched Linear Bandits

Advisor: Prof. [Pan Xu](#) (Department of Computer Science, Duke University)

Duke University

Aug. 2023 – Jan. 2024

- A theoretical work on **batched linear bandits**: devised the first algorithm to simultaneously achieve **asymptotic and non-asymptotic regret optimality** with the **least possible batch complexity**, a goal previously unattained.
- Confirmed the algorithm's superiority over existing baselines through rigorous experimentation.

Community Service

Conference Reviewer: NeurIPS, ICML, ICLR, RLC, ACL.

Skills

Programming: Python, PyTorch, CUDA, NKI (Trainium kernels), Lean 4, R, LaTeX, MATLAB, Mathematica.